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Homework 3

Non-Programming Exercises

All artifacts for Homework 3 can be found at:

https://github.com/juliansoh/EE524/tree/main/AU26_GPUCompute/Homework-3

https://github.com/juliansoh/EE524/tree/main/AU26_GPUCompute/Homework-3-MMULT

Questions I–III

I. Maximizing resident threads per SM

Answer: C — 512 threads per block.

This is a question of maximizing threads. With 128 threads per block, using all four blocks gives 512 threads. With 256 threads per block, all four blocks give 1,024 threads. With 512 threads per block, four blocks would give 2,048 threads, exceeding the limit of 1,536; three blocks give exactly 1,536 threads. With 1,024 threads per block, only one block can be deployed without exceeding the thread limit.

Threads per block	Resident blocks	Total threads
128	4	512
256	4	1,024
512	**3**	**1,536**
1,024	1	1,024

II. Threads in the vector-addition grid

Answer: C — 2,048 threads.

If each thread calculates one element of a vector of length 2,000, at least 2,000 threads are required. Blocks contain 512 threads and must be allocated in their entirety:

$$\text{Blocks required} = \left\lceil \frac{2000}{512} \right\rceil = 4, \qquad \text{Grid threads} = 4(512) = \boxed{2048}.$$

III. Warps diverging at the vector boundary

Answer: A — one warp.

Each warp contains 32 threads:

$$\frac{2000}{32} = 62.5 \text{ warps needed}, \qquad \frac{2048}{32} = 64 \text{ warps provisioned}.$$

The first 62 warps calculate $62(32) = 1,984$ elements. These warps do not diverge because all their threads perform the same vector calculation.

The remaining $2000 - 1984 = 16$ elements are handled by half of the 32 threads in warp 63. Half of that warp calculates and half does not, so this warp diverges.

All threads in warp 64 are unnecessary and do not calculate at all. That warp does not diverge. Therefore, the number of divergent warps is **one**.

Questions IV–V

IV. Feasible block assignments and occupancy

Answer: A, B, C, D, and E — all assignments are possible.

None of the assignments exceeds the maximum of 64 blocks or 2,048 threads per SM. Occupancy is calculated using

$$\text{Occupancy} = \frac{\text{Active warps}}{\text{Maximum warps}}, \qquad \text{Maximum warps} = \frac{2048}{32} = 64.$$

	Blocks	Threads/block	Total threads	Warps	Occupancy
A	8	128	1,024	32	50%
B	16	64	1,024	32	50%
C	32	32	1,024	32	50%
D	64	32	2,048	64	100%
E	32	64	2,048	64	100%

V. Resource limits and maximum occupancy

Answer: A is thread-limited, B is block-limited, and C is register-limited.

The system limits are 32 blocks, 2,048 threads, 65,536 registers, and 64 warps per SM.

	Blocks	Threads	Registers	Warps	Max. occupancy
A	16	2,048	61,440	64	100%
B	32	1,024	30,720	32	50%
C	7	1,792	60,860	56	87.5%

A: 128 threads per block, 30 registers per thread. More than 16 blocks would exceed the thread limit. With 16 blocks, there are 2,048 threads and 64 active warps, giving full occupancy.

B: 32 threads per block, 29 registers per thread. The 32-block limit prevents assigning more blocks even though more threads could be accommodated. There are 1,024 threads and 32 active warps, giving 50% occupancy.

C: 256 threads per block, 34 registers per thread. The register budget permits at most

$$\frac{65536}{34} \approx 1927 \text{ threads.}$$

Eight blocks would require $8(256) = 2,048$ threads, exceeding that register-limited thread count. Only seven blocks, or 1,792 threads, can be assigned. The resulting 56 active warps give

$$\text{Maximum occupancy} = \frac{56}{64} = \boxed{87.5\%}.$$

Numerical table entries are retained as supplied in the source responses.

Analysis: Control Divergence

Tiled MMULT — tile setup and block behavior

Tile/block setup and phases

Original handwritten work: Appendix A.1.

Assume T is both the tile width and tile height, so tiles are square. The tile size is T^2, with one thread block per tile, T^2 threads per block, and $T^2/32$ warps per block.

The matrix width is N. In one phase, a thread block can load only T columns, so the number of phases needed to load all columns is $\lceil N/T \rceil$. The output grid is $\lceil M/T \rceil \times \lceil N/T \rceil$.

Type 1: tiles are within bounds until the last tile. **Type 2:** tiles are partially out of bounds for the entire phase. The original block diagram is included in Appendix A.1.

Control divergence in loading Type 1 blocks

Original handwritten work: Appendix A.2.

Type 1 excludes the bottom block row. Each block contains $T^2/32$ warps, each executing $\lceil N/T \rceil$ phases:

$$\begin{aligned}
\text{Type 1 blocks} &= \left\lfloor \left\lceil \frac{M}{T} \right\rceil - 1 \right\rfloor \left\lceil \frac{N}{T} \right\rceil, \\
\text{Warp-phases per block} &= \frac{T^2}{32} \left\lceil \frac{N}{T} \right\rceil, \\
\text{Total Type 1 warp-phases} &= \frac{T^2}{32} \left\lceil \frac{N}{T} \right\rceil \left\lfloor \left\lceil \frac{M}{T} \right\rceil - 1 \right\rfloor \left\lceil \frac{N}{T} \right\rceil \\
&= \frac{T^2}{32} \left\lfloor \left\lceil \frac{N}{T} \right\rceil \right\rfloor^2 \left\lfloor \left\lceil \frac{M}{T} \right\rceil - 1 \right\rfloor.
\end{aligned}$$

The warps have control divergence only in their last phase. Let D_1 be the number of Type 1 warp-phases with control divergence:

$$D_1 = \begin{cases} \left\lfloor \left\lceil \frac{M}{T} \right\rceil - 1 \right\rfloor \left\lceil \frac{N}{T} \right\rceil \frac{T^2}{32}, & N \bmod T \neq 0, \\ 0, & N \bmod T = 0. \end{cases}$$

Control divergence in Type 2 blocks

Original handwritten work: Appendix A.3.

Type 2 consists of all blocks assigned to the bottom row:

$$\begin{aligned}
\text{Type 2 blocks} &= \left\lceil \frac{N}{T} \right\rceil, & \text{Total warps} &= \left\lceil \frac{N}{T} \right\rceil \frac{T^2}{32}, \\
\text{Phases per block } p &= \left\lceil \frac{N}{T} \right\rceil, & \text{Total Type 2 warp-phases} &= \left\lfloor \left\lceil \frac{N}{T} \right\rceil \right\rfloor^2 \frac{T^2}{32}.
\end{aligned}$$

The Type 2 control-divergent warp-phase count is

$$\boxed{D_2 = \left\lfloor \left\lceil \frac{N}{T} \right\rceil \right\rfloor^2 \frac{T^2}{32}}.$$

Analysis: Combined Impact

Total control divergence

Original handwritten work: Appendix A.3.

The total number of control-divergent warp-phases is the sum of the Type 1 and Type 2 counts:

$$D_{\text{total}} = D_1 + D_2.$$

Across the output grid, the total number of warp-phases and the combined impact are

$$\text{Total warp-phases} = \left\lceil \frac{M}{T} \right\rceil \left\lceil \left\lceil \frac{N}{T} \right\rceil \right\rceil^2 \frac{T^2}{32}, \qquad \text{Impact} = \frac{D_{\text{total}}}{\text{Total warp-phases}}.$$

Numerical check

Original handwritten work: Appendix A.4.

For $M = 100$, $N = 100$, and $T = 16$:

$$\left\lceil \frac{M}{T} \right\rceil = 7, \qquad \left\lceil \frac{N}{T} \right\rceil = 7, \qquad \frac{T^2}{32} = \frac{16^2}{32} = 8, \qquad p = 7.$$

$$\begin{aligned} D_1 &= (7-1)(7)(8) = 6(7)(8) = 336, \\ D_2 &= 7^2(8) = 392, \\ D_{\text{total}} &= 336 + 392 = 728, \\ \text{Total warp-phases} &= 7(7^2)(8) = 2744, \\ \text{Impact} &= \frac{728}{2744} = 0.2653 = \boxed{26.53\%}. \end{aligned}$$

When warp-divergence impact becomes minimal

Warp-divergence impact is minimal when boundary/tail conditions disappear or become negligible. This happens in either of these cases:

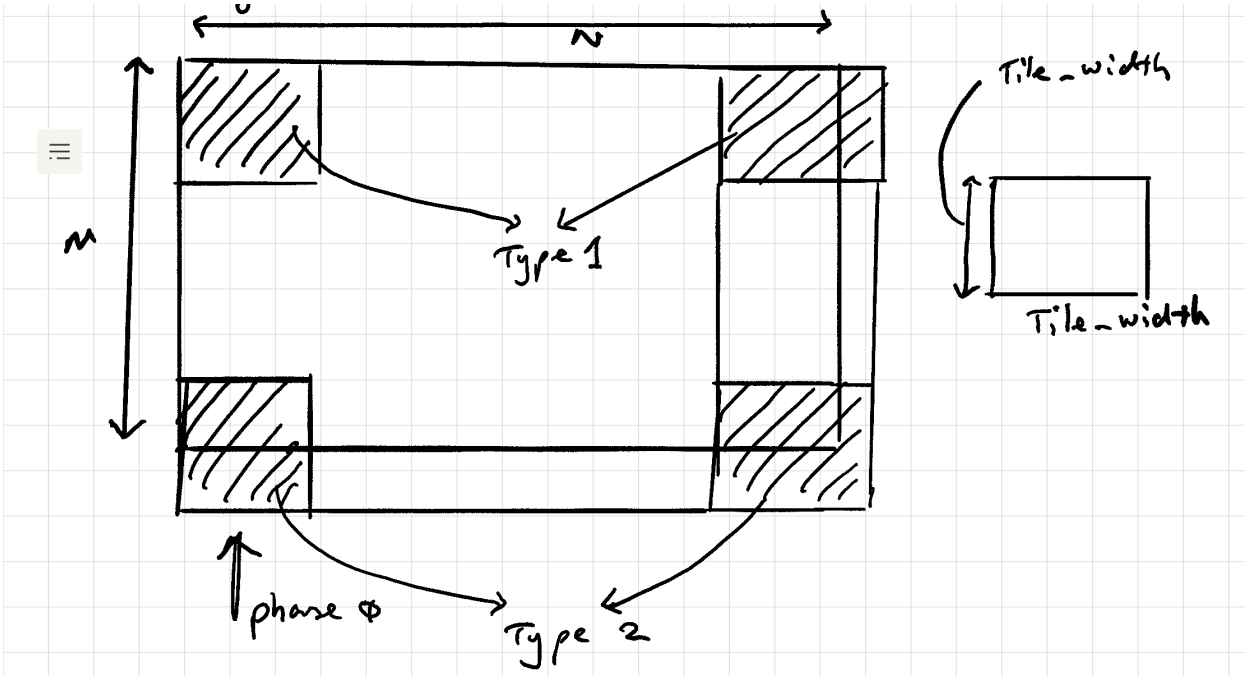
- **Exact tiling:** dimensions are multiples of the tile size ($m \bmod T = 0$ and $n \bmod T = 0$), so there are no partial edge/tail phases.
- **Large matrices relative to T:** edge blocks are a small fraction of all blocks, so divergence overhead is diluted.

Effect of tile size on arithmetic intensity

In tiled GEMM, arithmetic intensity increases roughly linearly with T. Larger tiles generally improve the compute-to-memory ratio, until limited by shared-memory/register pressure and occupancy.

Appendix A.1 Original handwritten analysis and calculations

Tile/block diagram, square-tile setup, and phase count.



Type 1 : where Tiles are within bounds until last Tile.

Type 2 : Tiles partially out of bounds for the entire phase.

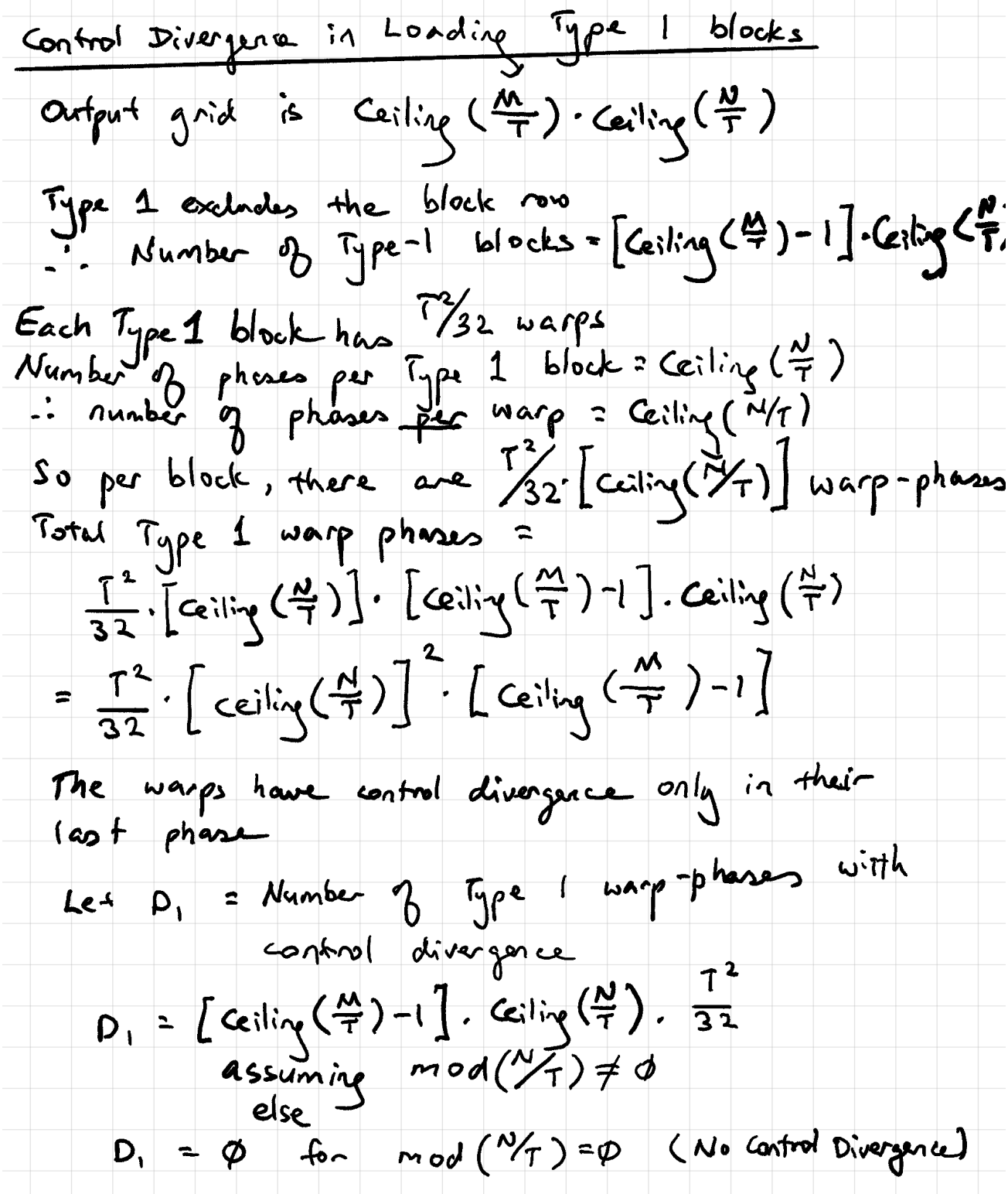
Assume T as the Tile_width and Tile_height (Square Tile)

1) Tile/block setup.
   - Tile size = T^2
   - 1 thread block per tile ⇒ T^2 threads per block
   - Warps per block = $T^2/32$

2) Phases
   - Width of matrix = N
   - In one phase, a thread block can only load T columns (size of tile)
   - Number of phases to load all columns = Ceiling$(\frac{N}{T})$

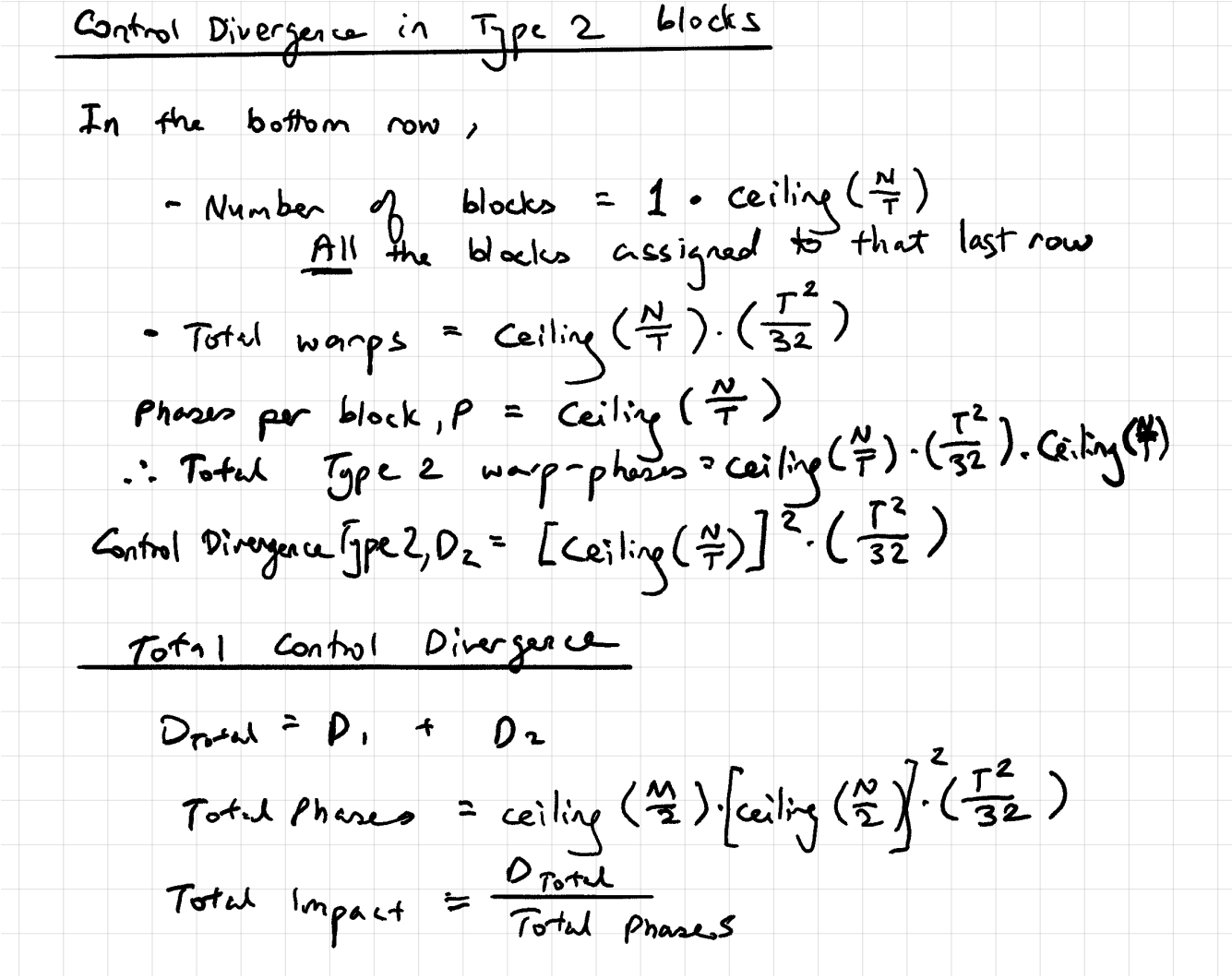
Appendix A.2 Original handwritten analysis and calculations

Type 1 blocks: warp-phase counts and final-phase divergence.



Appendix A.3 Original handwritten analysis and calculations

Type 2 blocks and total control-divergence expressions.



Appendix A.4 Original handwritten analysis and calculations

Numerical check for $M = N = 100$ and $T = 16$.

Numeric check

$M = 100$, $N = 100$, $T = 16$

$\text{Ceiling}\left(\frac{M}{T}\right) = 7$, $\text{ceiling}\left(\frac{N}{T}\right) = 7$

$\text{Warps} = \left(\frac{16^2}{32}\right) = 8$

$P = 7$

$D_1 = \left[\text{ceiling}\left(\frac{M}{T}\right) - 1\right] \cdot \text{ceiling}\left(\frac{N}{T}\right) \cdot \frac{T^2}{32}$

$= 6 \cdot 7 \cdot 8$

$= 336$

$D_2 = \left[\text{ceiling}\left(\frac{N}{T}\right)\right]^2 \cdot \left(\frac{T^2}{32}\right)$

$= 7^2 \cdot 8$

$= 392$

$D_{\text{Total}} = D_1 + D_2 = 336 + 392 = 728$

$\text{Total phases} = \text{ceiling}\left(\frac{M}{2}\right) \cdot \left[\text{ceiling}\left(\frac{N}{2}\right)\right]^2 \cdot \left(\frac{T^2}{32}\right)$

$= 7 \cdot 7^2 \cdot 8$

$= 2744$

$\text{Impact} = D_{\text{Total}} / \text{Total phases} = 728/2744$

$= 0.2653$

or

26.53%